PRODUCT X1234567 | Hold Issue Parametric Yield Investigation & Test Analysis

18 December, 1990

Confidentiality-preserving summary — anonymized identifiers retained from source review.

Project Overview

Objective: Understand the marginal parametric failure and final-test yield decline observed in the affected production lot. Investigate root contributors and determine appropriate disposition and corrective action pathway.

Approach

Compare test setup, verify repeatability, rescreen rejected units, review historical yield data, and analyze probe and final-test trends.

Scope

Elevated-temperature testing, reference correlation, in-line quality verification, and failure investigation across affected and reference lots.

Outcome

Lot contained pending technical review and disposition. No approved release or root cause determination at time of report.

18 December, 1990

Product Overview & Observed Issue

A production lot of Product X123456.7 experienced a marginal parametric failure during elevated-temperature testing. The primary failure category measured slightly below its lower specification limit. Final-step yield was 51.1% versus the 65% target — a 13.9 percentage-point gap triggering lot containment and investigation.

Actual Yield

Final-test yield result: 51.1% — 13.9pp below the 65% yield limit threshold

Target Yield

Yield target set at 65%. Gap of 13.9pp triggered lot containment and review.

51.1%

Test Yield: 51.1%

65%

Target: 65%

Product Information Table

Product X: X123456.7 | ASSY-LOTID: X1234567 | ASSY-SITE: MARS
FAB-LOTID: X1234567.1–.4 | FAB-SITE: MARS | PRB-LOTID: X123456.7
PRB-SITE: EARTH | LDT: 2777 | YIELD LIMIT: 65%

18 December, 1990

Confidentiality-preserving summary — anonymized identifiers retained from source review.

Repeatability Check Results:

Repeatability checks using the test setup, known-good units, and rejected units demonstrated consistent and reproducible results across all insertions. No test-setup variability was identified as a primary contributor to yield loss.

Repeatability & Force-Through Verification

Force-Through Verification:

Force-through verification confirmed that rejected units failed specifically at the affected parameter(s). Units continued to pass all subsequent test parameters, isolating the failure to a single parametric condition and ruling out broader device failure.

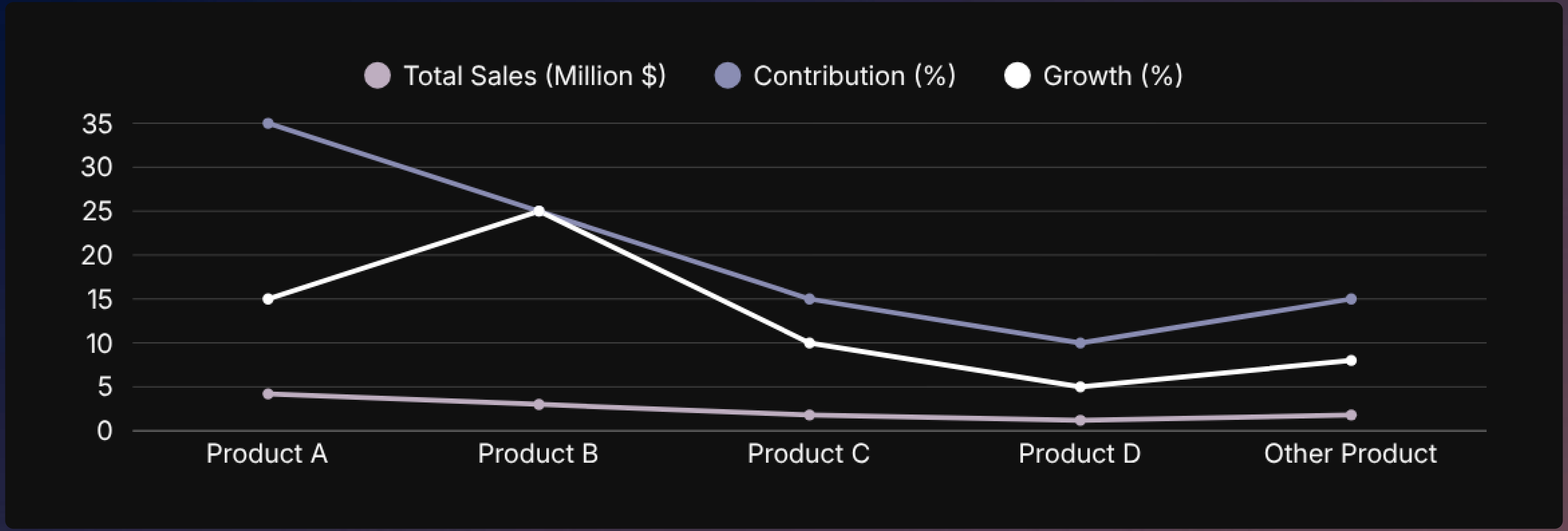
18 December, 1990

Confidentiality-preserving summary — anonymized identifiers retained from source review.

Reference Test Setup Data

25°C & 85°C

Test correlation performed at 25°C and 85°C using verified high-yield test hardware and supporting equipment.



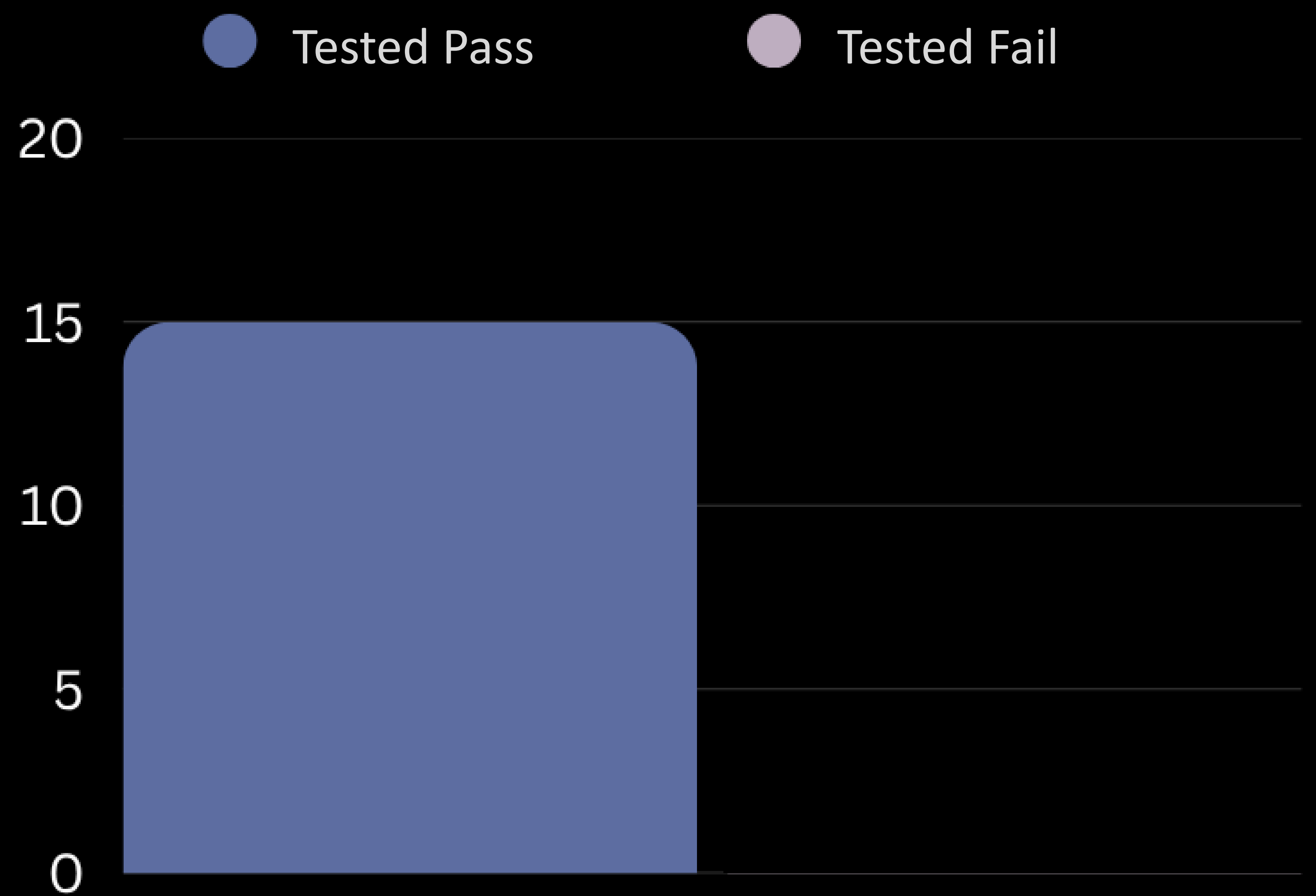
2 Fail Cats.

Initial data showed intermittent failures across two failure categories. Test-setup isolation and corrective action requested before reject rescreening.

18 December, 1990

Confidentiality-preserving summary — anonymized identifiers retained from source review.

Post-Test Quality Verification



100%

Post-Test Verification Yield Result

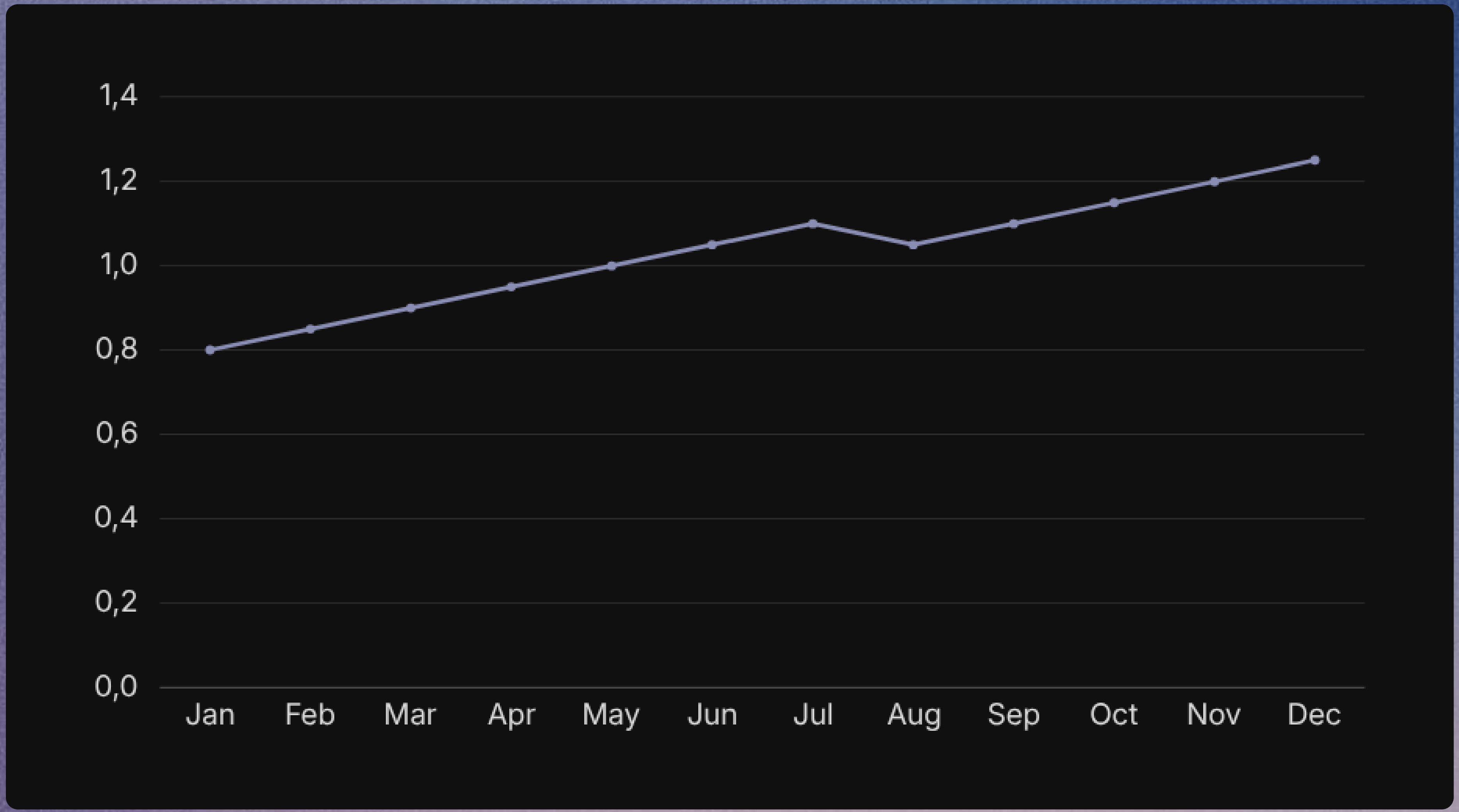
+VE

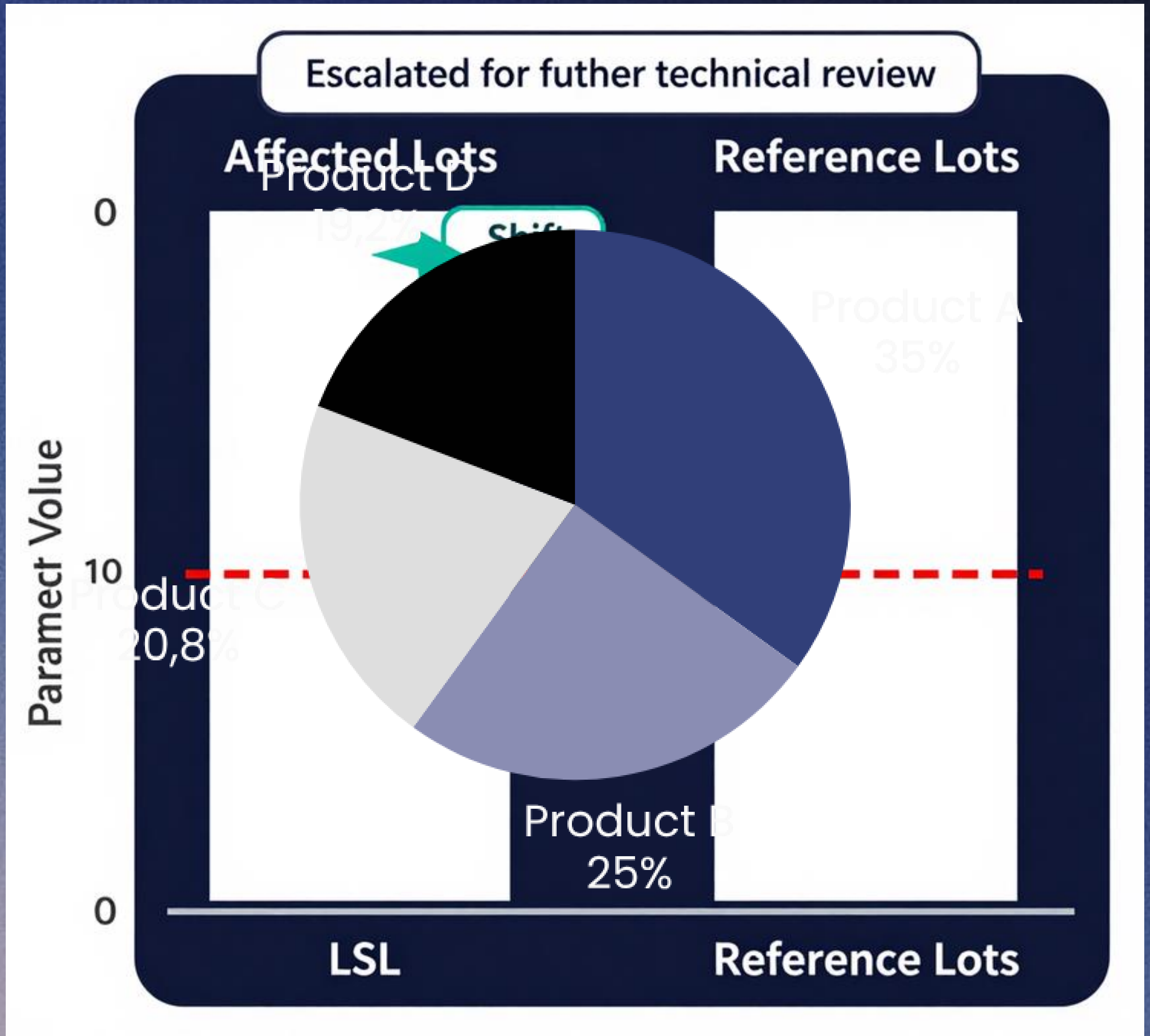
Quality Check: Positive Result

18 December, 1990

Confidentiality-preserving summary — anonymized identifiers retained from source review.

Probe-Test Performance: Affected vs. Reference Lots

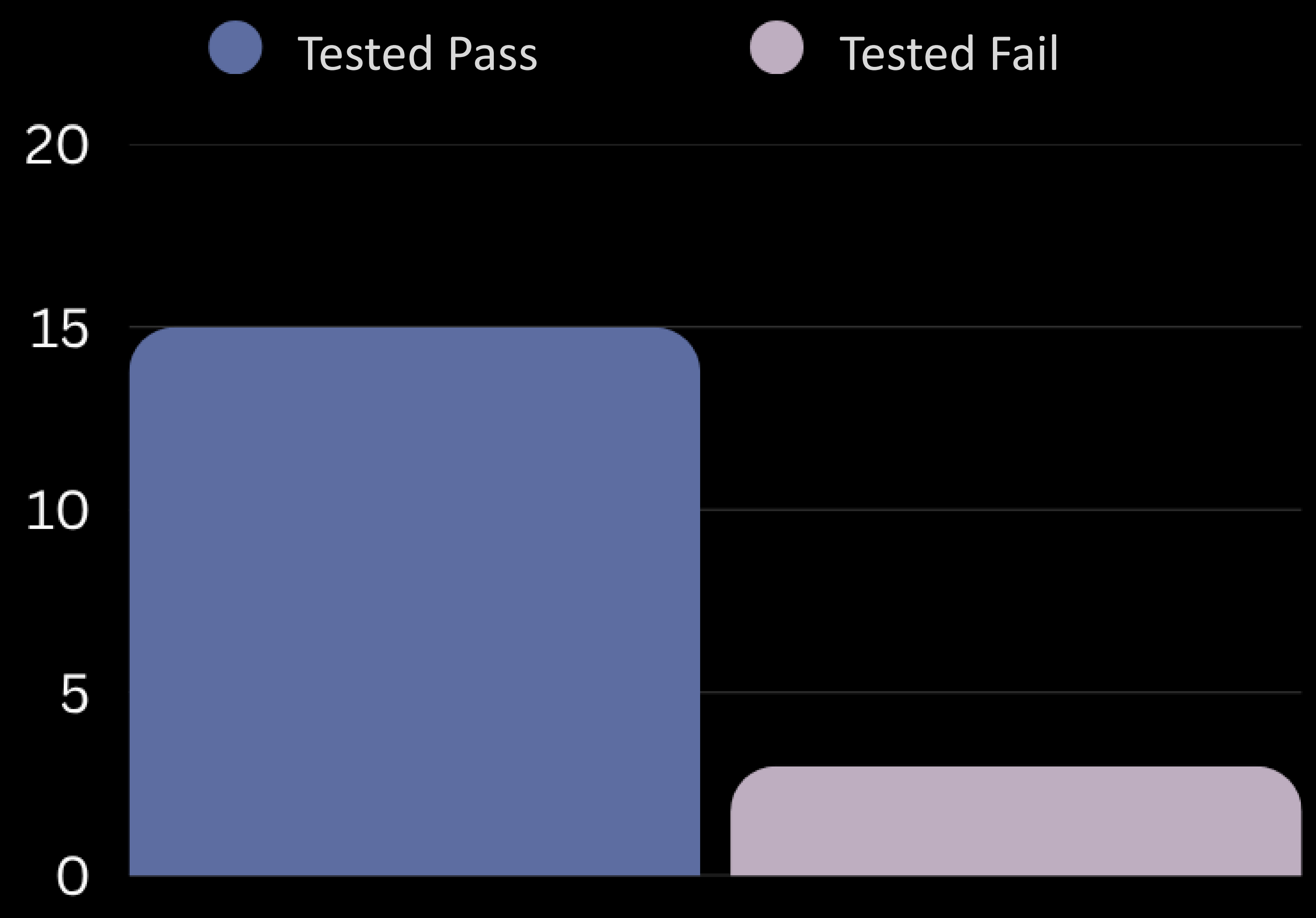




18 December, 1990

Confidentiality-preserving summary — anonymized identifiers retained from source review.

Final-Test Yield Trend



51.1%

Actual Final-Test Yield (vs. 65% target)
Affected lot — escalated for technical review

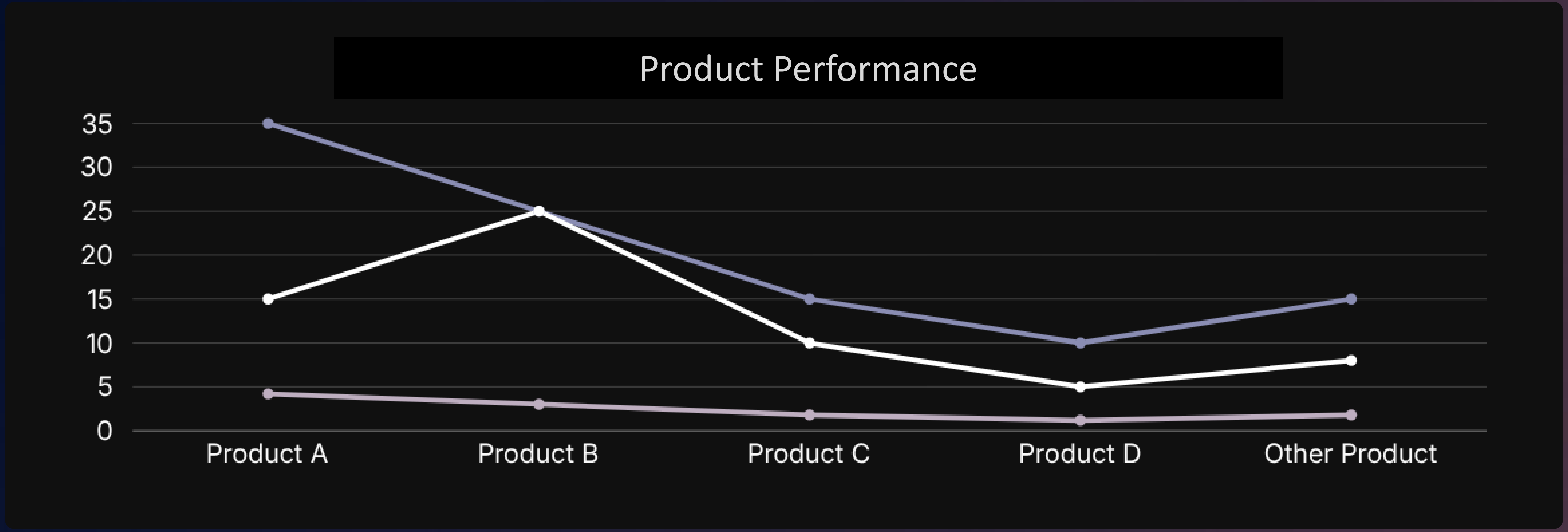
−13.9%

Gap vs. 65% yield limit
Parametric shift vs. reference lots

Historical Yield Trend

RECURRENCE

Historical data identified prior occurrences of same failure mode — supports process/product investigation



INSTABILITY

Multiple periods of yield instability observed in historical trend; pattern consistent with current failure category

18 December, 1990

Confidentiality-preserving summary — anonymized identifiers retained from source review.

Risk Evaluation & Disposition

Containment

Lot contained following significant final-test yield decline (51.1% vs. 65% target) and increased parametric fallout at affected parameter.

Status

UNDER CONTAINMENT — PENDING TECHNICAL REVIEW

The affected lot was contained following a significant final-test yield decline and increased parametric fallout. All available evidence has been reviewed: test setup repeatability was consistent, failures were isolated to the affected parameter, in-line quality verification passed at 100%, and probe data showed a shift toward the lower specification limit.

HOLD

Disposition: Pending

TBD

Corrective Action: Pending Determination

Evidence Summary

Test setup verification: consistent
Rejected units: failed at affected parameter, passed subsequent steps
PTQV result: 100% pass
Probe data: noticeable shift toward LSL

18 December, 1990

Confidentiality-preserving summary — anonymized identifiers retained from source review.

Key Findings:

Evidence did not support attributing failures to the test setup alone. The probe-data shift toward the lower specification limit is a potential contributor. Historical recurrence of the same failure mode supports continued technical review of process- or product-related contributors.

Containment Status:

Lot remains under containment pending risk evaluation, final disposition, and corrective action. No root cause has been confirmed. All findings escalated for further technical review before release decision.

Investigation Conclusion

18 December, 1990

Confidentiality-preserving summary — anonymized identifiers retained from source review.

Next Steps & Reporting Notes

Action Items:
- Complete technical review of probe-data shift
- Evaluate process & product contributors
- Finish test-setup isolation & corrective action
- Determine lot disposition
- Define corrective & preventive actions (CAPA)
- Continue monitoring final-test yield & affected parameter

Reporting Note: Embedded source charts were low-resolution; exact unreadable values were not reproduced. Trend visuals are qualitative unless values were explicitly stated in the source (verified: 51.1% actual yield; 65% target).

18 December, 1990